

Table 1: Ablation of SemanticLink components across multiple random seeds

Model Variant	Seed	Accuracy	Precision	Recall	F1-score	ROC-AUC	Avg Precision
Raw text	42	0.9227	0.8603	0.8243	0.8419	0.9592	0.8843
	123	0.9300	0.8468	0.8787	0.8624	0.9662	0.8796
	7	0.9415	0.8588	0.9163	0.8866	0.9666	0.8750
	21	0.9342	0.8492	0.8954	0.8717	0.9629	0.8812
	99	0.9310	0.8619	0.8619	0.8619	0.9622	0.8697
	555	0.9289	0.8462	0.8745	0.8601	0.9618	0.8841
	Mean $\pm$ Std	0.9314 $\pm$ 0.0057	0.8539 $\pm$ 0.0066	0.8752 $\pm$ 0.0285	0.8641 $\pm$ 0.0134	0.9632 $\pm$ 0.0026	0.8790 $\pm$ 0.0052
SRL Markup	42	0.9342	0.8520	0.8912	0.8712	0.9687	0.8892
	123	0.9331	0.8302	0.9205	0.8730	0.9632	0.8686
	7	0.9363	0.8371	0.9247	0.8787	0.9649	0.8815
	21	0.9363	0.8504	0.9038	0.8763	0.9700	0.8972
	99	0.9321	0.8372	0.9038	0.8692	0.9651	0.9005
	555	0.9342	0.8411	0.9079	0.8732	0.9610	0.8747
	Mean $\pm$ Std	0.9343 $\pm$ 0.0015	0.8413 $\pm$ 0.0077	0.9086 $\pm$ 0.0112	0.8736 $\pm$ 0.0031	0.9655 $\pm$ 0.0031	0.8853 $\pm$ 0.0115
Without Contrastive Loss	42	0.9342	0.8577	0.8828	0.8701	0.9622	0.8600
	123	0.9310	0.8264	0.9163	0.8690	0.9599	0.8634
	7	0.9279	0.8400	0.8787	0.8589	0.9630	0.8750
	21	0.9331	0.8302	0.9205	0.8730	0.9585	0.8438
	99	0.9289	0.8577	0.8577	0.8577	0.9689	0.8934
	555	0.9279	0.8455	0.8703	0.8577	0.9643	0.8577
	Mean $\pm$ Std	0.9305 $\pm$ 0.0025	0.8429 $\pm$ 0.0122	0.8877 $\pm$ 0.0231	0.8644 $\pm$ 0.0064	0.9628 $\pm$ 0.0034	0.8655 $\pm$ 0.0155
Without Cross Attention	42	0.9300	0.8554	0.8661	0.8607	0.9698	0.8994
	123	0.9331	0.8405	0.9038	0.8710	0.9671	0.8908
	7	0.9342	0.8667	0.8703	0.8685	0.9674	0.8988
	21	0.9383	0.8571	0.9038	0.8798	0.9660	0.8988
	99	0.9289	0.8406	0.8828	0.8612	0.9666	0.8830
	555	0.9289	0.8519	0.8661	0.8589	0.9646	0.8723
	Mean $\pm$ Std	0.9323 $\pm$ 0.0034	0.8520 $\pm$ 0.0093	0.8821 $\pm$ 0.0163	0.8667 $\pm$ 0.0073	0.9669 $\pm$ 0.0016	0.8905 $\pm$ 0.0100

Table 2: Baseline comparison without SemanticLink components across multiple random seeds

Baseline Model	Seed	Accuracy	Precision	Recall	F1-score	ROC-AUC	Avg Precision
SecRoBERTa	42	0.9342	0.8729	0.8619	0.8674	0.9734	0.9104
	123	0.9237	0.8705	0.8159	0.8423	0.9684	0.8959
	7	0.9269	0.8789	0.8201	0.8485	0.9676	0.9094
	21	0.9143	0.8792	0.7615	0.8161	0.9640	0.9105
	99	0.9154	0.8726	0.7741	0.8204	0.9581	0.8959
	555	0.9310	0.8745	0.8452	0.8596	0.9709	0.9019
	Mean $\pm$ Std	0.9242 $\pm$ 0.0074	0.8748 $\pm$ 0.0032	0.8131 $\pm$ 0.0357	0.8424 $\pm$ 0.0188	0.9671 $\pm$ 0.0049	0.9040 $\pm$ 0.0064
SecBERT	42	0.9300	0.8839	0.8285	0.8553	0.9672	0.8986
	123	0.9310	0.8745	0.8452	0.8596	0.9727	0.8998
	7	0.9289	0.8638	0.8494	0.8565	0.9696	0.9059
	21	0.9237	0.8517	0.8410	0.8463	0.9662	0.8805
	99	0.9258	0.8652	0.8326	0.8486	0.9683	0.8926
	555	0.9300	0.8644	0.8536	0.8589	0.9683	0.9015
	Mean $\pm$ Std	0.9282 $\pm$ 0.0026	0.8673 $\pm$ 0.0100	0.8417 $\pm$ 0.0088	0.8542 $\pm$ 0.0050	0.9687 $\pm$ 0.0021	0.8965 $\pm$ 0.0082
SBERT	42	0.9394	0.8787	0.8787	0.8787	0.9689	0.9065
	123	0.9363	0.8771	0.8661	0.8716	0.9689	0.9017
	7	0.9363	0.9083	0.8285	0.8665	0.9730	0.9088
	21	0.9216	0.9059	0.7657	0.8299	0.9553	0.8997
	99	0.9300	0.9135	0.7950	0.8501	0.9667	0.9134
	555	0.9394	0.8987	0.8536	0.8755	0.9712	0.9181
	Mean $\pm$ Std	0.9338 $\pm$ 0.0063	0.8970 $\pm$ 0.0142	0.8312 $\pm$ 0.0400	0.8621 $\pm$ 0.0170	0.9673 $\pm$ 0.0057	0.9080 $\pm$ 0.0064
RoBERTa-Large	42	0.9279	0.8373	0.8828	0.8595	0.9539	0.8620
	123	0.9279	0.8400	0.8787	0.8589	0.9604	0.8667
	7	0.9279	0.8295	0.8954	0.8612	0.9510	0.8629
	21	0.9300	0.8282	0.9079	0.8663	0.9602	0.8461
	99	0.9300	0.8468	0.8787	0.8624	0.9639	0.8681
	555	0.9363	0.8450	0.9121	0.8773	0.9647	0.8918
	Mean $\pm$ Std	0.9300 $\pm$ 0.0030	0.8378 $\pm$ 0.0070	0.8926 $\pm$ 0.0136	0.8642 $\pm$ 0.0063	0.9590 $\pm$ 0.0050	0.8663 $\pm$ 0.0135

Table 3: SemanticLink trained with different encoder backbones

Encoder	Seed	Accuracy	Precision	Recall	F1-score	ROC-AUC	Avg Precision
SecRoBERTa	42	0.9310	0.8650	0.8577	0.8613	0.9609	0.9013
	123	0.9331	0.8788	0.8494	0.8638	0.9539	0.8899
	7	0.9310	0.8502	0.8787	0.8642	0.9632	0.8895
	21	0.9300	0.8613	0.8577	0.8595	0.9624	0.9028
	99	0.9258	0.8590	0.8410	0.8499	0.9605	0.8892
	555	0.9289	0.8548	0.8619	0.8583	0.9633	0.8987
	Mean $\pm$ Std	0.9300 $\pm$ 0.0023	0.8615 $\pm$ 0.0090	0.8577 $\pm$ 0.0116	0.8595 $\pm$ 0.0048	0.9607 $\pm$ 0.0032	0.8952 $\pm$ 0.0059
SecBERT	42	0.9300	0.8707	0.8452	0.8577	0.9626	0.8898
	123	0.9300	0.8440	0.8828	0.8630	0.9653	0.8876
	7	0.9321	0.8686	0.8577	0.8632	0.9660	0.8883
	21	0.9321	0.8750	0.8494	0.8620	0.9653	0.8892
	99	0.9342	0.8697	0.8661	0.8679	0.9609	0.8901
	555	0.9310	0.8681	0.8536	0.8608	0.9643	0.9016
	Mean $\pm$ Std	0.9316 $\pm$ 0.0014	0.8660 $\pm$ 0.0101	0.8591 $\pm$ 0.0125	0.8624 $\pm$ 0.0031	0.9641 $\pm$ 0.0018	0.8911 $\pm$ 0.0048
Sentence-BERT	42	0.9373	0.8566	0.8996	0.8776	0.9673	0.9023
	123	0.9373	0.8683	0.8828	0.8755	0.9662	0.8915
	7	0.9258	0.8529	0.8494	0.8512	0.9694	0.8855
	21	0.9363	0.8904	0.8494	0.8694	0.9649	0.8960
	99	0.9394	0.8819	0.8745	0.8782	0.9608	0.9071
	555	0.9310	0.8811	0.8368	0.8584	0.9653	0.8892
	Mean $\pm$ Std	0.9345 $\pm$ 0.0047	0.8718 $\pm$ 0.0137	0.8654 $\pm$ 0.0219	0.8684 $\pm$ 0.0102	0.9657 $\pm$ 0.0026	0.8953 $\pm$ 0.0075